

SPCX

A working primer: what the words mean, what we measured, and what it actually showed.

Written for someone learning to trade. Every term is explained the first time it appears, and again in the glossary at the back. Nothing here is advice about what to buy — it is a record of what ninety days of SPCX price data said when we asked it careful questions.

24 September 2026 · 70 trading sessions, 12 June to 22 September · 1,040 signals · 27,164 one-minute bars

[Start here](#)

What this document is

You asked me to explain the findings in terms you could follow, and to teach the vocabulary so you can think and talk like a trader. This is that. It is in two halves.

The first half teaches the language. A price chart, the two indicators we use, and the handful of measurements that decide whether a trading idea is any good. If you understand the second half of this document, you will understand most of what traders say to each other.

The second half is what the data said. Four findings, three of them negative. I have put them in plainly, because a tool that tells you what you hoped to hear is worth less than nothing.

One honest note before we start. The main idea we tested — a signal built from something called the MACD — did not work. It has now failed three separate tests. That is not a failure of the tools; it is the tools doing their job. Most trading ideas do not work, and the ones that look best on a chart are usually the ones that fool you hardest. Finding that out from ninety days of data is enormously cheaper than finding it out from ninety days of trading.

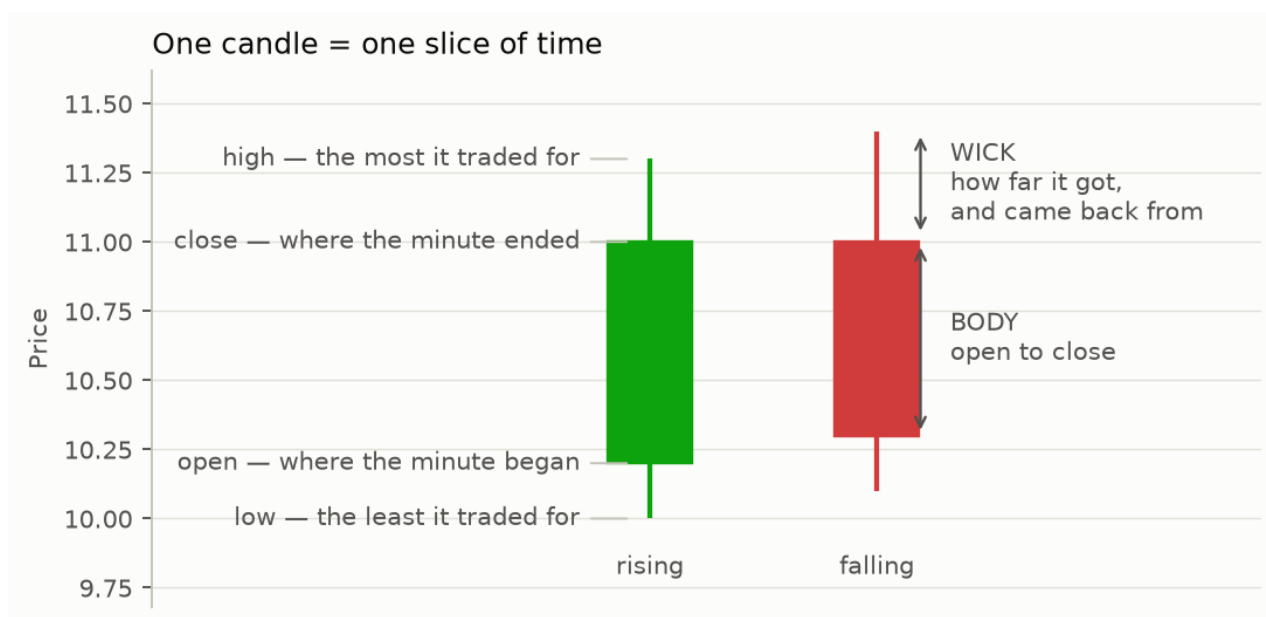
The one thing worth taking away

Of the trades that eventually worked, the typical one dipped only **0.27%** against you first. A stop placed **0.75%** below your entry would have kept you in **88%** of them. That is a real, usable number, and it came out of the losing study. Section 9 explains it.

1. How to read a price chart

A **bar** — or **candle**, the same thing drawn prettier — is a summary of one slice of time. We use one-minute candles, so each one answers four questions about a single minute: where it started, where it ended, the highest anyone paid, and the lowest.

Those four numbers have names, and traders use them constantly: the **open**, the **close**, the **high** and the **low**. You will see them written as OHLC.



Green when the close is above the open, red when it is below. That colour convention is universal — every platform you will ever use draws it this way.

The **body** is the thick part, from open to close. The **wicks** (also called shadows or tails) are the thin lines showing how far price got before coming back. A long upper wick means buyers pushed the price up and then lost it again within that minute — often a sign that sellers were waiting up there.

You saw this on 18 September: the 09:30 candle ran up more than two dollars and closed 98 cents off its high. That 98 cents was a wick, and the six red candles that followed suggested it had been telling the truth.

Volume

Volume is how many shares changed hands. It is the second dimension of every chart and the one beginners ignore. Price tells you where; volume tells you how much conviction was behind it. A one-percent move on tiny volume and a one-percent move on enormous volume are different events, even though the chart line looks identical.

Raw volume on its own is nearly useless, because volume is wildly uneven through the day — the first minutes after the open routinely carry fifty times what a quiet minute at lunchtime does. So we always compare a minute against **the same clock minute on previous days**. That is what your alerts mean when they say *1.8× usual for 09:35*: nearly twice the shares that 09:35 normally carries.

Why the comparison has to be time-of-day

An earlier version of this idea compared each minute against a rolling average of the morning so far. It looked reasonable and was badly wrong: it turned the question *is this minute unusually busy?* into *is it near the open yet?*, because the open is always busier than what came before it. Comparing 09:35 against previous 09:35s is the only honest version.

The session

The **regular session** runs 09:30 to 16:00 Eastern. Before and after that is **pre-market** and **after-hours** (together, **extended hours**) — trading still happens, but thinly, and the rules are different. Your tools start reading at 09:25 deliberately, to show you the last five minutes of positioning before the bell.

Part one — the language

2. VWAP: the day's average price

VWAP stands for volume-weighted average price. It answers: across everyone who has traded this stock today, what is the average price they paid?

Weighted by volume means a trade of ten thousand shares counts ten times as much as a trade of a thousand. So VWAP is not the average of the price line; it is the average of the *money*.

Traders watch it for a simple reason: it is a scoreboard. If you bought this morning and price is above VWAP, you are doing better than the average buyer today. Large institutions are often measured against VWAP explicitly — which means they have a real incentive to buy under it and sell over it, and that makes it a line where things happen.

One detail that matters, and that we got right

VWAP **resets every morning**. It is the average for *this day*, not a rolling average of the last few hundred minutes. Get that wrong and your line drifts further from the real one every day you run — and it stops being the line anyone else is looking at, which was the entire point of drawing it.

Part one — the language

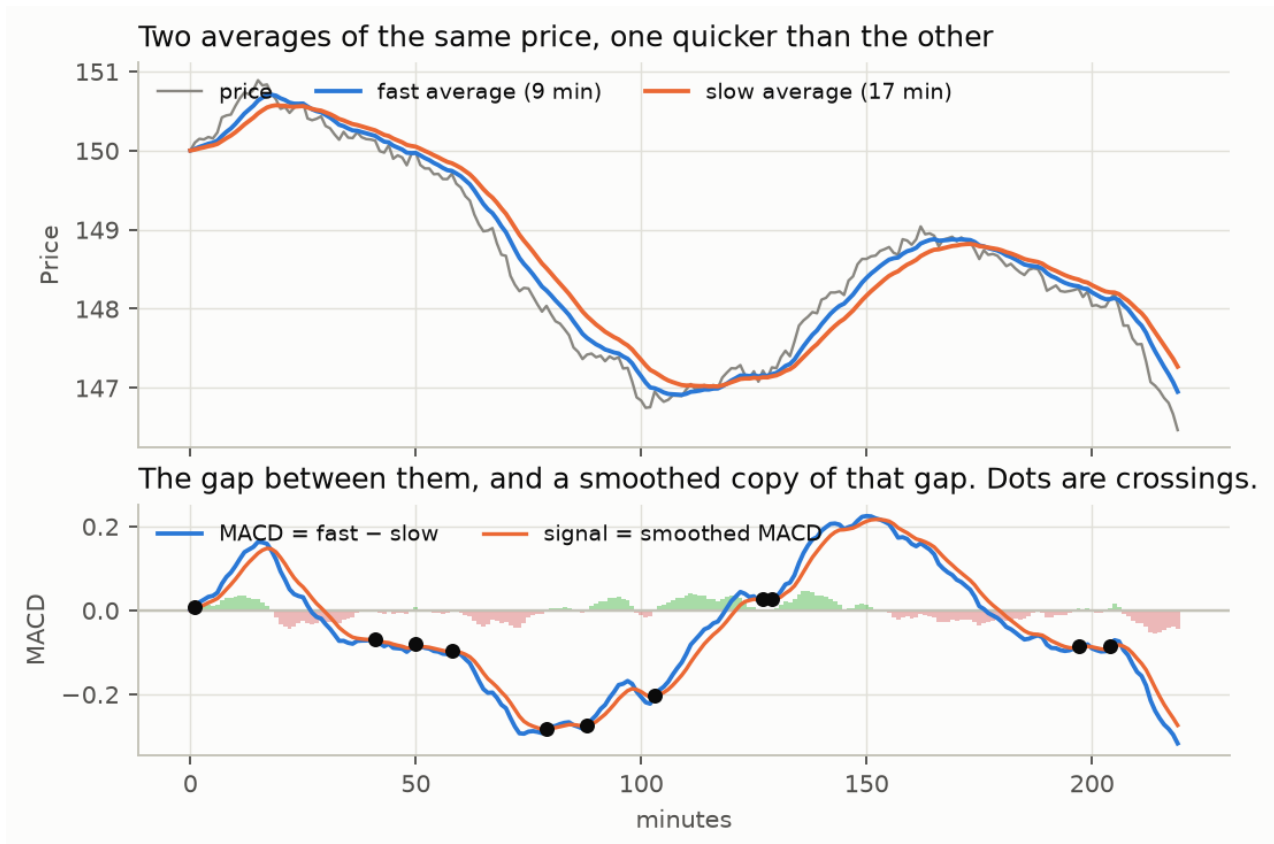
3. The MACD

The **MACD** — moving average convergence divergence, a name nobody enjoys — is built in three steps, and each one is simple.

Step one: two averages. Take the average price over the last 9 minutes, and the average over the last 17. Both follow the price, but the 9-minute one turns faster because it has less history weighing it down.

Step two: subtract them. Fast minus slow. That difference is the MACD line. When price accelerates upward the fast average pulls away from the slow one and the line rises. When price stalls, they converge and it falls back toward zero. Above zero means the short term is running ahead of the longer term; below zero, behind.

Step three: smooth it. Take a 6-minute average of the MACD line itself. That is the **signal line**. When the MACD line crosses up through it, the recent trend has turned up faster than its own recent average — that is a **crossover**, and it is what your alert fires on.



Top: price and its two averages. Bottom: the gap between them (the MACD line), a smoothed copy of that gap (the signal line), and the difference between those two drawn as bars — the histogram. The dots mark crossovers.

The numbers 9, 17 and 6 are your settings. The conventional ones are 12, 26 and 9. There is no magic in either set, and this matters more than it sounds: **the periods are counted in bars, not minutes**. Your 9/17/6 on one-minute bars means 9, 17 and 6 minutes. The exact same settings on five-minute bars would mean 45, 85 and 30 minutes — a completely different indicator wearing the same name.

Why the chart in your PDF uses one-minute bars

Your report draws five-minute candles, but the MACD panel underneath is computed on one-minute bars. That is deliberate. If it were recomputed on the five-minute candles it would be a plausible-looking line with no relationship to your alerts, and its crossovers would not sit under your signal markers. Same function, same numbers, same line the alert read.

Divergence

You used the word **divergence** once, and it is worth knowing you meant something we have not built. Divergence is when price makes a new low but the MACD makes a *higher* low — the selling is continuing but with less force behind it each time. It is a genuine and more advanced idea. What your alert actually fires on is the crossover described above.

4. How you judge whether a signal is any good

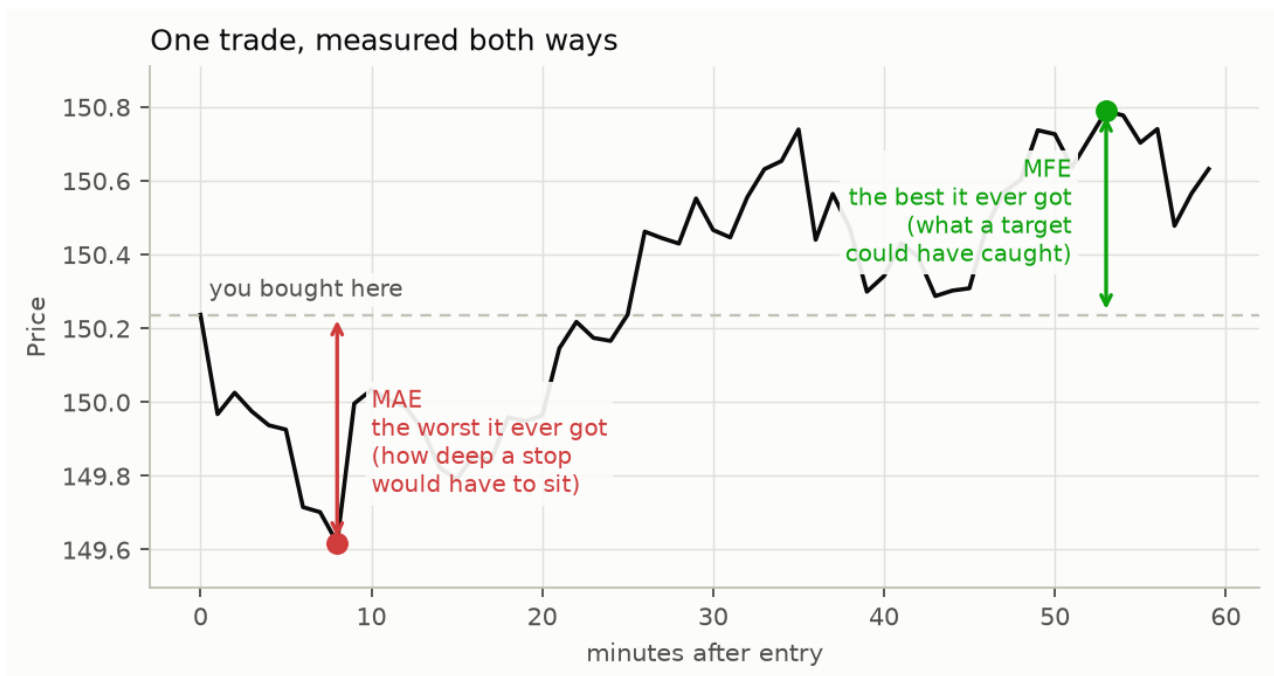
This is the section that matters most, and it is the vocabulary you were missing. Everything in the second half of this document is built from these five ideas.

MFE and MAE

Suppose you buy at 150.23. Over the next hour price wanders: down to 149.62, then up to 150.80, then back to 150.63 where you are looking at it.

MFE — maximum favourable excursion — is the best it ever got. Here, 150.80, or +0.38%. It is the ceiling on what any target could have caught. Set a target above the typical MFE and it simply never fills.

MAE — maximum adverse excursion — is the worst it ever got. Here, 149.62, or -0.41%. It is how deep a stop would have needed to sit to survive this trade. Set a stop inside the typical MAE and you get thrown out of trades that were about to work.



The same trade, measured both ways. MFE and MAE are simply the best and worst points reached — they say nothing about where it finished.

These two numbers, measured across a thousand trades, are how you size a stop and a target from evidence rather than from a round number that feels about right.

Stop, target, bracket

A **stop** (stop-loss) is an order that sells automatically if price falls to a level you set. A **target** (or limit sell) sells automatically if price rises to a level you set. Together they form a **bracket**: your trade now has a floor and a ceiling, and neither requires you to be watching.

This is the one that cost you money

A stop you intended to place is not a stop. The whole value of a resting stop order is that it works while your attention is elsewhere — at your job, asleep, or on a phone with no signal. Most brokers let you attach the stop and target to the entry so all three go in as one action. If forgetting is possible, it will eventually happen; make it impossible instead of resolving to remember.

Edge, spread, and slippage

An **edge** is the amount by which a method beats doing nothing thoughtful. It is always a comparison, never an absolute — and picking what you compare against is where most self-deception happens.

The **spread** is the gap between the highest price anyone is currently willing to pay and the lowest anyone will sell for. You buy at the higher one and sell at the lower one, so the spread is a cost you pay on every round trip whether or not your broker charges commission. **Slippage** is the extra you lose when your order fills at a slightly worse price than you saw — normal when things are moving fast.

Together these set a floor. On a \$150 stock, a penny or two of spread is roughly 0.01%. So an edge of 0.004% per trade is not a small edge; it is **nothing at all**, buried inside a cost you cannot avoid. This is why the studies keep saying an edge under 0.05% is noise.

The control

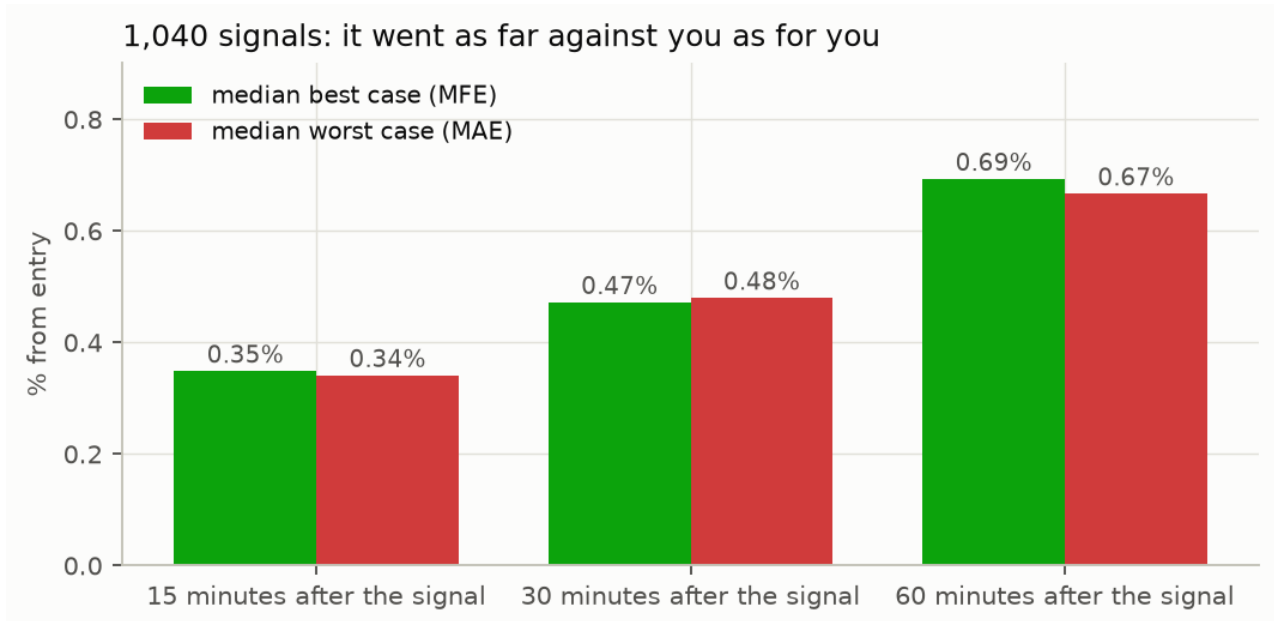
A trading idea can only be judged against an alternative. Ours is deliberately stupid: **buy every fifth bar, regardless of anything**. No signal, no thought, no conditions.

This is the single most important piece of machinery in the whole project. A stock that drifts upward over ninety days will make almost any buying rule look profitable — including one that ignores the chart entirely. Without the dumb version to compare against, you have no way to tell whether your clever idea contributed anything, or whether you simply measured the drift and called it a strategy.

If your method cannot beat buying at random, then whatever it is detecting, it is not something that makes money.

5. The signal is a coin flip

Across 70 sessions the MACD crossover fired 1,040 times — about 15 a day. For each one we measured how far price went in each direction over the next 15, 30 and 60 minutes.



Median best case against median worst case. If the signal found anything, the green bars would stand taller than the red ones.

They are the same height. At every horizon, the typical signal went almost exactly as far against you as it went for you. An hour later, price was higher **51%** of the time — which is a coin flip with a rounding error.

This pattern has a name: it is what a **random walk** looks like. A random walk is a price that moves by an unpredictable amount each minute, with no memory of what it just did. You cannot forecast one, and every measurement you take of one comes back symmetric — exactly as these did.

It also loses to the dumb version

We then tried every combination of stop and target — six stops against six targets, thirty-six brackets — on the signals, and the same thirty-six on the buy-every-fifth-bar control.

	Best bracket found	Average per trade
On the MACD signals	0.75% stop / 3.00% target	+0.107%
Buying at random	1.00% stop / 3.00% target	+0.111%
The signal's edge		−0.004%

The signal lost to entering at random. And the +0.107% flatters it, because it is the winner of thirty-six attempts on the same data — the best of thirty-six coin-flip experiments always looks better than it is.

This is now the third time we have asked, on 29 days, then 69, then 90. It has failed every time. The first test produced a 54% figure that looked promising; it did not survive more data. That is the normal life cycle of a trading idea, and recognising it early is the skill.

What this does not mean

It does not mean the alert is useless to *you*. These studies measure the signal as an autopilot — buy every time it fires, with a fixed bracket, no judgement. You are not doing that. You use it to decide where to look, and then you decide. What the studies say is narrower and still important: **do not take a trade because the MACD fired**. Take it because of what you see when you get there.

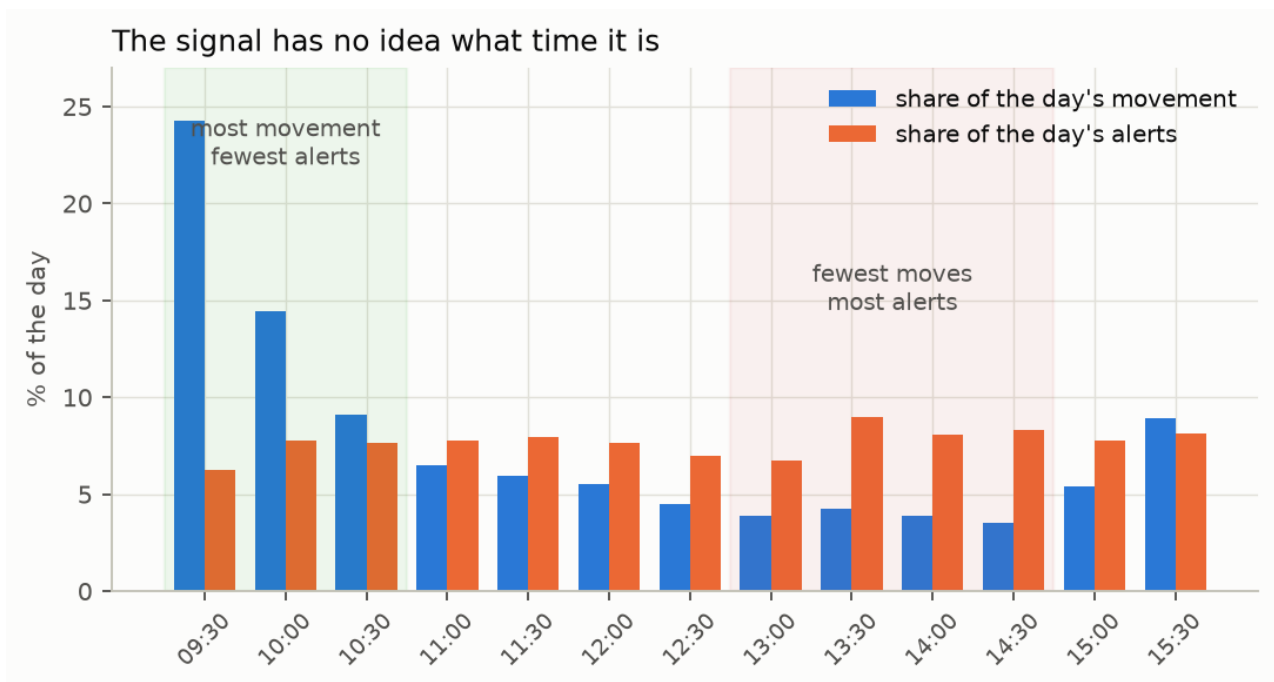
Part two — what the data said

6. The signal has no idea what time it is

This is the finding that changed how your tools behave, and the only structural thing we found.

SPCX does not move evenly through the day. The average one-minute candle spans 0.585% of price in the half hour after the open, and 0.181% around 14:30 — the stock is three times livelier in the morning. Every trader knows this in the abstract; here it is measured on your stock.

The MACD does not know it. It fires at a flat rate all day.



Blue: where the day's movement actually is. Orange: where your alerts arrive. They are almost mirror images.

Stretch	Share of movement	Share of alerts
09:30–11:00	48%	22%
13:00–14:30	16%	32%

The 13:30 half hour produces more alerts than any other slot in the day — 94 of them — during the stretch where the stock moves least. Those are the ones pulling you away from your job to look at nothing.

Worse, the afternoon signals were not merely fewer-moving but worse: 13:00 through 14:30 is four consecutive half hours where the typical signal went *further against you than for you*. One bad slot would be noise. Four

adjacent ones agreeing is a pattern.

What we changed

Your alerts now run **09:40 to 11:00** and go quiet after. That takes you from about 15 alerts a session to about 3, while keeping well over a third of the day's movement in view.

Everything is still measured and written to the database after 11:00 — the gate decides what buzzes your phone, never what gets recorded. If the data later says the window is wrong, we will have the data to say so.

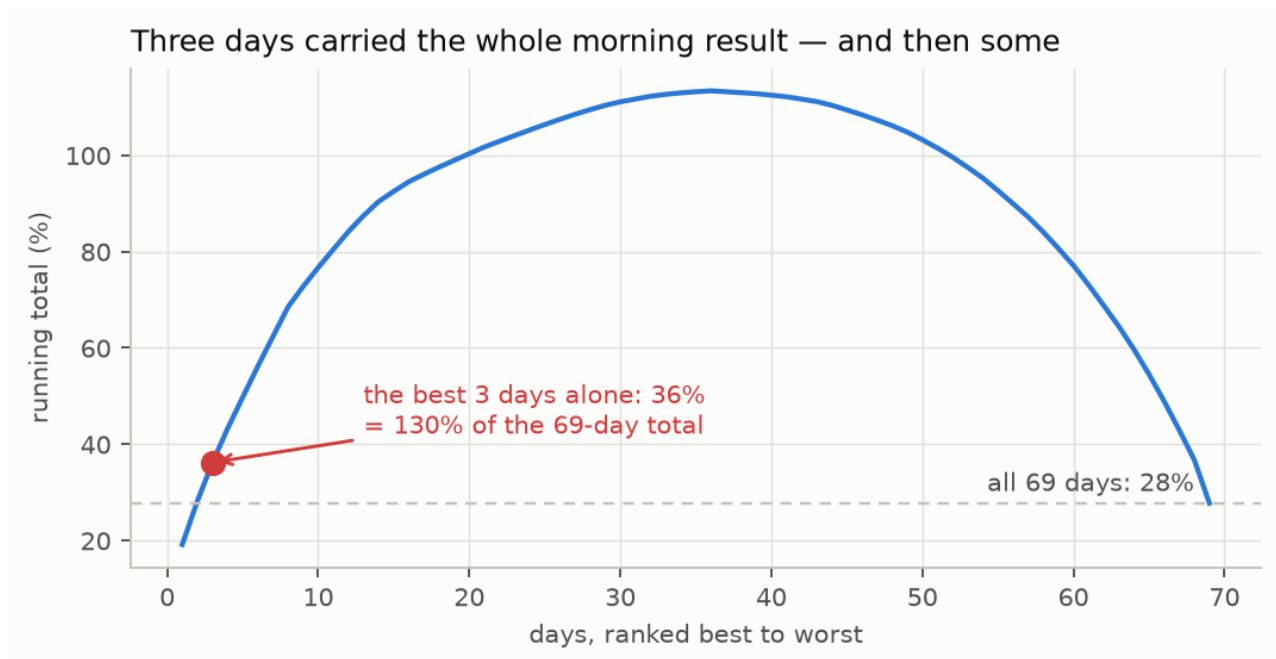
Be clear about what this buys

A quieter phone, not a better entry. We tested the gate with the control restricted to the same hours — because narrowing to the busiest part of the day raises returns by itself, and without that control we would have measured the time of day and called it a signal. With the control in place, no window beat random. Three alerts you read beat fifteen you learn to ignore; that is the entire argument, and it does not need an edge to be true.

Part two — what the data said

7. Three days out of sixty-nine

When we restricted to the morning window, one measurement came back faintly positive. Before getting attached to it, we asked a question you should always ask of a good-looking result: **where did it come from?**



Every morning-window day, ranked best to worst, added up one at a time. A healthy result climbs steadily. This one is carried by its first three days and gives most of it back.

Adding up every morning signal's one-hour return across 69 days gives +27.8%. The best three days alone give +36.2% — **130% of the total**. Remove 28 July, 8 September and 30 June and the whole thing is negative.

The median day contributed +0.23%. Fifty-two per cent of days were positive. That is a coin flip with three lucky Tuesdays on top.

This is **concentration**, and checking for it is one of the most useful habits you can build. An average is a single number that hides its own story. Two strategies can report the same average return where one made a little on most days and the other made nothing on 66 days and a fortune on three. The first is a strategy. The second is a lottery ticket you have already scratched.

The question to ask of any good result

If I remove the best three days, is it still there? If the answer is no, you have not found a method. You have found three days.

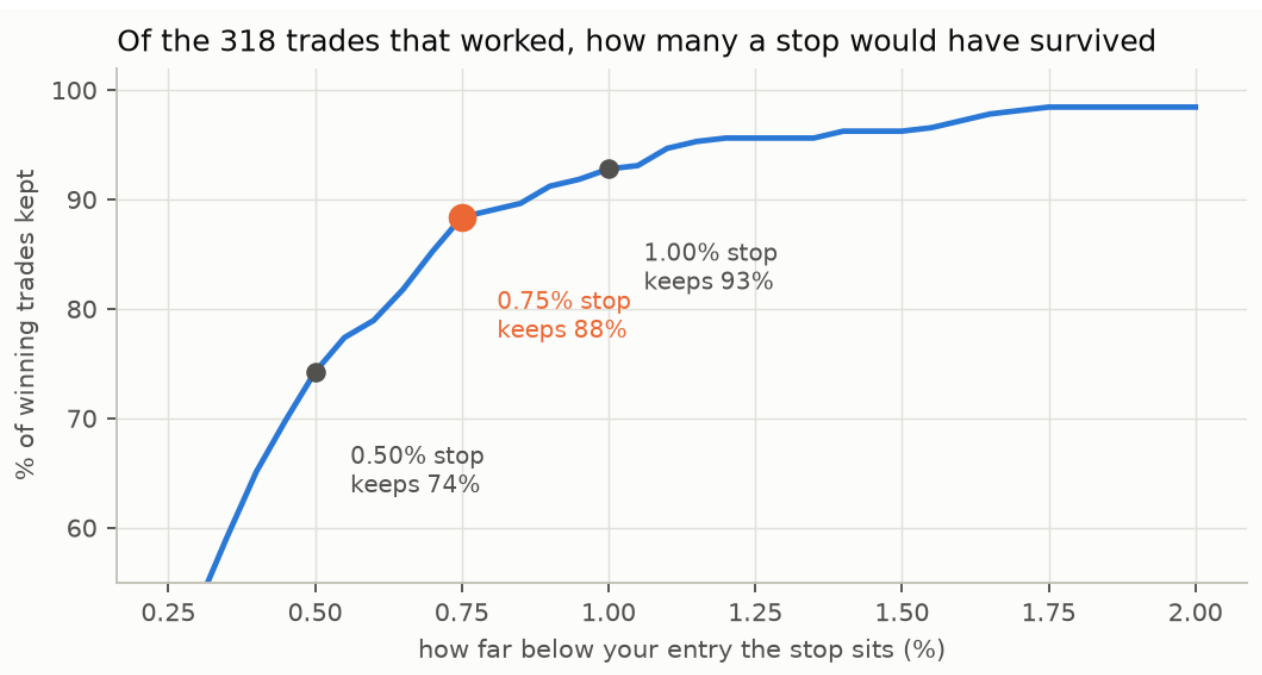
Part two — what the data said

8. Where the stop goes

This is the one finding you can act on tomorrow, and it corrects advice I gave you earlier in the week.

I had told you a 0.5% stop would take you out of half your trades, based on the median MAE across all 1,040 signals — about -0.48% . That number is true and it is the wrong number, because it is dominated by the losers. Losing trades dig deep by definition; including them tells you where *failures* go, not where winners go.

The right question is narrower: **of the trades that ended up working, how far did they dip first?** The answer is much more encouraging.



318 of the 1,040 signals finished at least 0.5% up an hour later. This is how many of those a stop at each distance would have survived.

Stop distance	Winners kept	In the 09:40–11:00 window
0.50%	74%	69%
0.75%	88%	87%
1.00%	93%	90%
1.50%	96%	94%

The median winner dipped only **0.27%** before going. Winners mostly just go — they do not typically spend twenty minutes underwater first. That is why the curve bends sharply and then flattens: past about 0.75% you are paying a lot more risk for very few extra winners.

The recommendation, and its limits

0.75% below your entry keeps you in 88% of the trades that work while cutting losers far sooner than 1% does. Then size the position so that a 0.75% loss is a dollar amount you are genuinely fine losing — because roughly half of all entries will hit it.

What this does **not** do is make the entries good. A well-placed stop on a coin-flip entry is still a coin flip; it just fails cheaply and predictably instead of expensively and at random. That is worth a great deal, and it is not an edge.

9. How not to fool yourself

Everything above rests on a handful of disciplines. They are the real skill in this work, they are learnable, and they are what separates a trader with a method from a trader with a story.

In-sample and out-of-sample

Data you have already looked at is **in-sample**. Data you have not is **out-of-sample**. Any rule invented by staring at in-sample data will fit it well, because you built it to. The only honest test is on data the rule has never seen.

This is the difference between *explaining* the past and *predicting* the future, and almost every failed trading system in existence confused the two.

Multiple comparisons

Test one idea at the 5% threshold and you have a one-in-twenty chance of a false positive. Test twenty ideas and you should *expect* one to look good by luck alone. Report only that one and you have produced a discovery out of nothing.

We searched thirty-six brackets in section 5. We looked at thirteen half-hour slots in section 6. If I told you the best of those was meaningful without mentioning how many I tried, I would be lying to you with true numbers.

Pre-registration

The fix is to write down the test **before** you run it: the rule, the bracket, what would count as success. Then run it once. If it fails, it failed — you do not get to adjust the rule until it passes.

We did this with the first study. The pre-registered claim was that 54% of signals would still be higher an hour later. On more data it came back 51%. It failed, and saying so was the entire value of having written it down.

A live example: the VWAP idea

Here is one working through the process right now, so you can watch the discipline being applied rather than just described.

In the per-signal data, morning signals that fired *above* VWAP looked better than those below: best-case to worst-case ratio of 1.67 against 0.79, and higher an hour later 58% of the time against 48%.

That is the most interesting thing in the file. It is also exactly the shape of the 54% that already fooled us once: about a hundred observations, one of four cells examined, and roughly 1.7 standard errors from a coin flip — a gap that appears by chance more often than people expect.

It does have a mechanism behind it, which is a point in its favour. VWAP is a line real institutions are measured against, so it is a place where behaviour genuinely changes. A pattern with a reason is more likely to be real than one without. But a plausible story is how you talk yourself into things, not how you test them.

So it gets tested properly, or not at all

The rule is fixed now: **only signals in 09:40–11:00 with price above VWAP, at a 0.75% stop and a 2.00% target**, against the same control restricted the same way. It will be run on trading days **before** 12 June 2026 — days neither of us has looked at.

If it survives that, it is worth something. If it does not, it joins the 54% and we say so and move on. What we will not do is search the existing ninety days for the version that looks best; that is not a test, it is a portrait.

Reference

10. Glossary

After-hours Trading between 16:00 and 20:00 Eastern. Thin, and prices can move a long way on very little.

Bar / candle One slice of time summarised as open, high, low, close and volume. We use one minute.

Baseline / control The deliberately unsophisticated alternative your idea must beat. Ours buys every fifth bar regardless.

Body The thick part of a candle: open to close.

Bracket An entry with a stop and a target attached, so the trade is bounded without you watching.

Concentration When a result comes from a handful of days rather than steadily. A warning sign, not a detail.

Crossover The MACD line crossing up through its signal line. What your alert fires on.

Divergence Price making a new low while the indicator makes a higher low — weakening pressure. Not what we built.

Edge How much a method beats the control by. Always a comparison. Under 0.05% a trade, it is inside the spread and therefore nothing.

EMA Exponential moving average: an average that weights recent prices more heavily. The MACD is built from two of them.

Extended hours Pre-market and after-hours together — outside 09:30–16:00 Eastern.

Fill The actual execution of your order, at the actual price you got.

Higher after The share of signals where price was higher N minutes later. 50% is a coin flip.

Histogram The bars on a MACD panel: the gap between the MACD line and its signal line.

IEX / SIP Two market data feeds. IEX is one exchange, free. SIP is the full consolidated tape from all exchanges. Never compare a number from one against a baseline from the other.

In-sample Data you have already examined. Any rule fits it, because you built the rule on it.

Limit order An order to trade at a set price or better. Will not fill at a worse price; may not fill at all.

MACD Fast average minus slow average, plus a smoothed copy of that difference. A momentum indicator.

MAE Maximum adverse excursion: the worst price reached before the trade closed. Sizes your stop.

MFE Maximum favourable excursion: the best price reached. Caps what a target can catch.

Market order An order to trade immediately at whatever price is available. Fast, and you pay the spread.

Out-of-sample Data the rule has never seen. The only honest test.

Pre-market Trading between 04:00 and 09:30 Eastern.

Pre-registration Writing down the test before running it, so a failure cannot be quietly renamed a success.

Random walk A price with no memory of its last move. Unpredictable by construction, and it measures symmetric — which is what ours did.

Relative volume This minute's volume against what that same clock minute usually carries. Never against a rolling average.

Session The regular trading day, 09:30 to 16:00 Eastern.

Signal line A smoothed average of the MACD line. The line it crosses.

Slippage The difference between the price you expected and the price you got.

Spread The gap between the best bid and the best offer. A cost you pay on every round trip, commission or not.

Stop / stop-loss A resting order that sells if price falls to your level. Works whether or not you are watching. That is the point.

Target A resting order that sells if price rises to your level.

Volume Shares traded. Price says where; volume says with how much conviction.

VWAP Volume-weighted average price for the day. Resets each morning. A scoreboard institutions are measured against.

Wick / shadow The thin lines above and below a candle body: how far price got and came back from.

Reference

11. Where things stand

	Status
MACD crossover as an entry signal	Failed three tests. Do not trade it on its own.
Time-of-day gate, 09:40–11:00	Shipped. Buys attention, not edge.
Stop at 0.75%	Supported by the data. Use it.
Volume spike alarm	Running. Threshold 1.5× is a starting guess, not a finding.
Morning candle stream	Running. Descriptive, and never advisory.
VWAP filter	Hypothesis. Pre-registered, not yet tested out of sample.
Your trades on the chart	Waiting on an export from your broker.

What the tools do

Three programs, all read-only. None of them can place an order; none of them knows your broker or your account. That was deliberate from the first day and has not changed.

Program	What it does
open_candles.py	Reads the tape to your phone. Full detail 09:25–10:00, then volume spikes only, to 16:00.
spcx_alert.py	Watches the MACD. Alerts 09:40–11:00; records all day.
daily_report.py	Builds the one-page PDF. Rebuilds every 15 minutes while the market is open.

Closing

Three of the four findings in this document are negative, and I want to be plain about why that is not a disappointing result.

You now know something about SPCX that most people trading it do not: that this particular signal, on this particular stock, is indistinguishable from chance, and that the busiest-looking part of the afternoon is where your attention is worth least. You learned it from ninety days of data rather than ninety days of losses. And you got one genuinely useful number out of it, which is where to put a stop.

The machinery that produced these answers is the durable part. The next idea you have can be put through the same mill — control, concentration check, pre-registration, out-of-sample test — in an afternoon. That is the thing worth having. Not this signal; the ability to find out.